

Generation Interconnection System Impact Study Report

For Queue Project Q17-0726

Carlton Hills 138 kV

March 02, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0726. The Interconnection Customer proposes a 77.0 MW solar photovoltaic facility interconnecting at the Carlton Hills 138 kV substation in the Virginia area. The request was received on January 15, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 220 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,888,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mecklenburg Power Sta to Person 345.0 kV	Bear to Clover 345.0 kV line	101.2	2,711	6.2
Clover to Bear 345.0 kV	Person to Mecklenburg Power Station 345.0 kV	100.9	2,871	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mecklenburg Power Station to Person 345.0 kV	Reconductor line	\$23,380,000	4% (\$1,062,000)
Clover to Bear 345.0 kV	Reconductor line	\$40,605,000	4% (\$1,826,000)
	Total allocated to project		\$2,888,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.